Recall

Work from memory, then check with your tutor.

R1. Find $\frac{1}{2} + \frac{1}{3}$.

Answer: _____

R2. Complete the ratio-table pair: (3, 4), (9, __).

Answer: _____

R3. 9 of 12 tiles are green. What percent are green?

Answer: _____

R4. How many $\frac{1}{4}$-meter pieces fit in 3 meters?

Answer: _____

S. Return here near the end of the lesson

I can check a fraction quotient by multiplying it by ___.

Fraction division and reciprocals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Share $\frac{3}{4}$ liter equally among 5 bottles. How much per bottle?

Answer: _____

P2. How many $\frac{3}{5}$-cup groups fit in 4 cups? Include a fraction of a group.

Answer: _____

P3. Find $\frac{7}{8} \div \frac{1}{4}$. Check by multiplying the quotient by $\frac{1}{4}$.

Answer: _____

Fraction division and reciprocals

Follow the slide labels. P = problem; F = your turn; D = discussion.

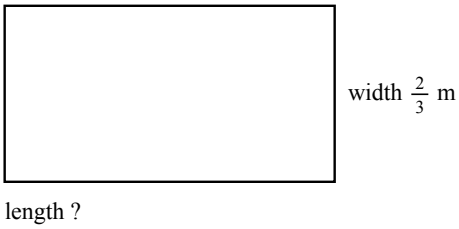
P4. Find $1\ \frac{7}{8} \div \frac{5}{8}$.

Answer: _____

P5. A cord is $3\ \frac{1}{2}$ meters long. Each piece is $\frac{7}{8}$ meter. How many pieces?

Answer: _____

P6. A rectangle has area $\frac{5}{6}$ square meter and width $\frac{2}{3}$ meter. Find its length.



Fraction division and reciprocals

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. A box has edges $\frac{2}{3}$, $\frac{3}{4}$, and 2 meters. Find its volume.

Answer: _____

F. Complete: $\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times __ = __$. Check by multiplying by $\frac{2}{3}$.

Answer: _____

D. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

Answer: _____

Mixed review

M1. Find $3 \times \frac{2}{3}$.

Answer: _____

M2. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

Answer: _____

M3. 3 markers cost $1.95. Find dollars per marker.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A ribbon is $2\frac{5}{8}$ yards long. Each piece is $\frac{3}{8}$ yard. How many pieces can be cut?

Answer: _____

Dividing fractions

The big idea

Division asks how many divisor-sized groups fit. Equal small units make the comparison visible. The reciprocal rule records that reasoning efficiently.

Words to know

Dividend: The amount being divided. Example: In $2 \div \frac{3}{4}$, the dividend is 2.

Divisor: The amount that names one group in a grouping problem. Example: In $2 \div \frac{3}{4}$, one group is $\frac{3}{4}$.

Quotient: The result of division. Example: $\frac{3}{4} \div \frac{1}{4} = 3$ groups.

Reciprocal: A number that multiplies with another number to make 1. Example: $\frac{3}{5}$ and $\frac{5}{3}$ are reciprocals.

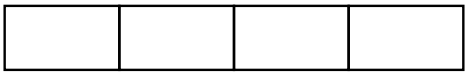
Mixed number: A whole number plus a fraction. Example: $2\ \frac{1}{3} = \frac{7}{3}$.

Volume: The amount of three-dimensional space a solid fills. Example: A box uses cubic units.

How it works

Five sixths is five fourths of a four-sixths group. The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.

one group: 4 sixths



amount: 5 sixths



One whole has $\frac{5}{3}$ groups of size $\frac{3}{5}$. Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.

one group: 3 fifths



one whole: 5 fifths



Worked examples

1. Find $\frac{3}{4} \div \frac{1}{2}$ with a tape.

Rename one half as two fourths.

Three fourths contains one full pair of fourths and one fourth left.

The remaining fourth is half of a two-fourths group.

Answer: $1\ \frac{1}{2}$

2. Find $\frac{2}{3} \div \frac{3}{5}$.

One whole contains $\frac{5}{3}$ groups of size $\frac{3}{5}$.

Two thirds of a whole contains $\frac{2}{3} \times \frac{5}{3}$ groups.

Multiply to get $\frac{10}{9}$, or $1\ \frac{1}{9}$.

Answer: $1\ \frac{1}{9}$

Common mistakes

Reverses dividend and divisor. Why: Has not named the available amount. Fix: Label amount first and group size second.

Multiplies the fractions. Why: Applies a rule before making sense of division. Fix: Count equal pieces before writing a rule.

Inverts the dividend. Why: Memorizes a rule without a division meaning. Fix: Check quotient × divisor = dividend.

Try it

1. How many $\frac{3}{4}$-liter groups fit in $\frac{7}{8}$ liter? Include part of a group.

2. A bottle is $\frac{3}{4}$ full and holds $\frac{2}{3}$ liter so far. Find its full capacity.

3. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

4. A ribbon is $4\ \frac{1}{2}$ meters long. Pieces measure $\frac{3}{8}$ meter. How many pieces fit?

5. Find $\frac{4}{7} \div \frac{2}{5}$. Check the result by multiplying by $\frac{2}{5}$.

Answers

1. $1\ \frac{1}{6}$ groups 2. $\frac{8}{9}$ liter 3. $\frac{5}{24}$ liter 4. 12 pieces 5. $1\ \frac{3}{7}$

Where this connects

This builds on fraction multiplication and equivalent ratios. Decimal division will also preserve group counts by renaming units.